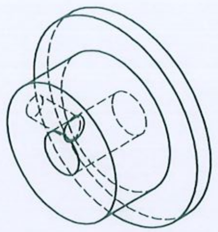
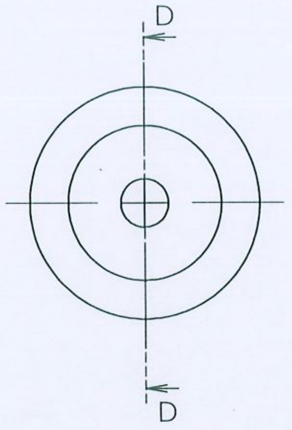
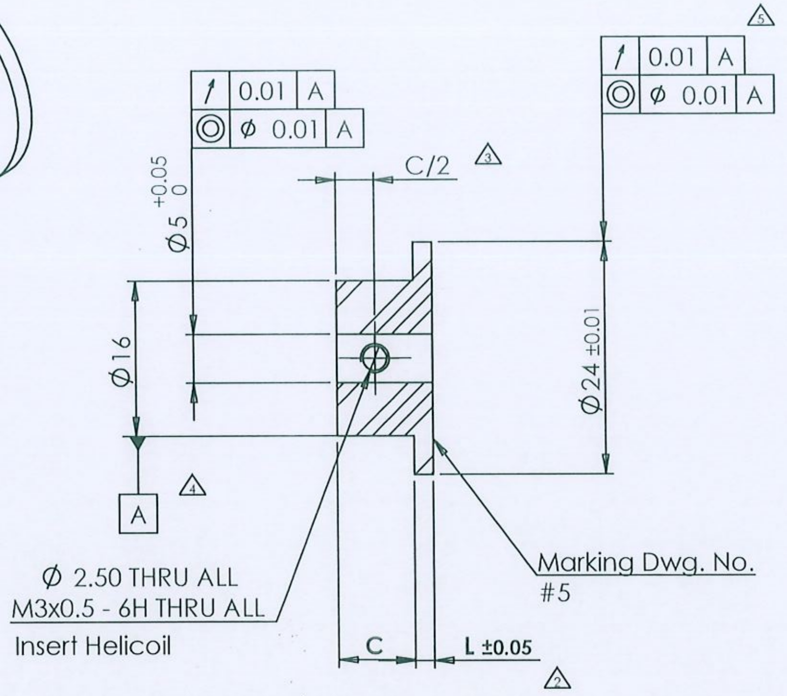
PART No. TGM-04-268-1-xx | REV. 5

Rt=6







Ø 2.50 THRU ALL
M3x0.5 - 6H THRU ALL
Insert Helicoil

Marking Dwg. No.
#5

SECTION D-D
SCALE 2 : 1.5

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

☑	Mat'l Tick by user
☑	Nylon blue
☐	Pom white
☐	Peek gray
☐	PP Gray

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ITEM	L	C
-01	2	10
-02	1.1	10
-03	2	6.4
-04	1.1	6.4
-05	2	4.5
-06	1.1	10
-07	2	10
-08	1.1	10
-09	2	6.4
-10	1.1	6.4
-11	2	4.5
-12	1.1	4.5
-13	2	3.6
14		
-15		

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	21 Jan'15	Preecha	Add: 2 ==> L Etc.; RQ.24/15
3	----	24 Aug'15	Preecha	Add: Table + C Etc.; RQ.374/15
4	----	27 Jan'16	Preecha	Change: Datam A
5	----	13 Nov'17	Preecha	Add: Geometric etc.;

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S
6.3S
25S
100S
~ UNCUT

M/C. Vis Sleeve | Q'ty 12
MATERIAL See table
FINISH -
HEAT TREATMENT -

List Request No. 2015

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 20 Oct' 14

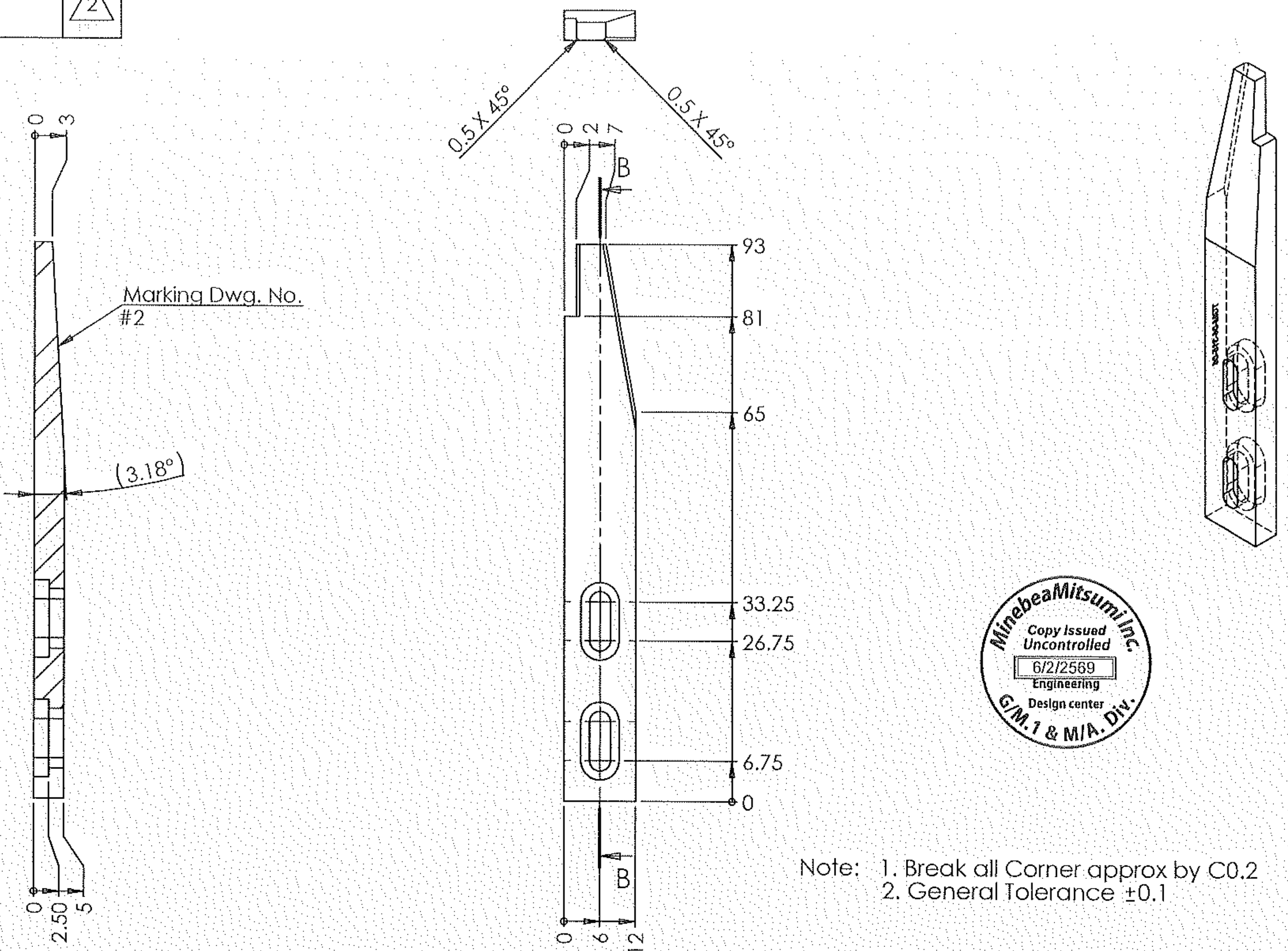
NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Roller (1)

PART No. TGM-04-268-1-xx | REV. 5

PART No. TGM-04-315-04 | 2

Rt=12



SECTION B-B

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G/M.1 & M/A. DIV.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	8 Aug'12	Preecha	Add: C'Bore

UNLESS SPECIFIED	
GENERAL TOLERANCE	
[illegible]	±0.05
[illegible]	±0.1
[illegible]	±0.2
[illegible]	±0.3
[illegible]	±0.4
[illegible]	±0.8
ANGLE	±1°

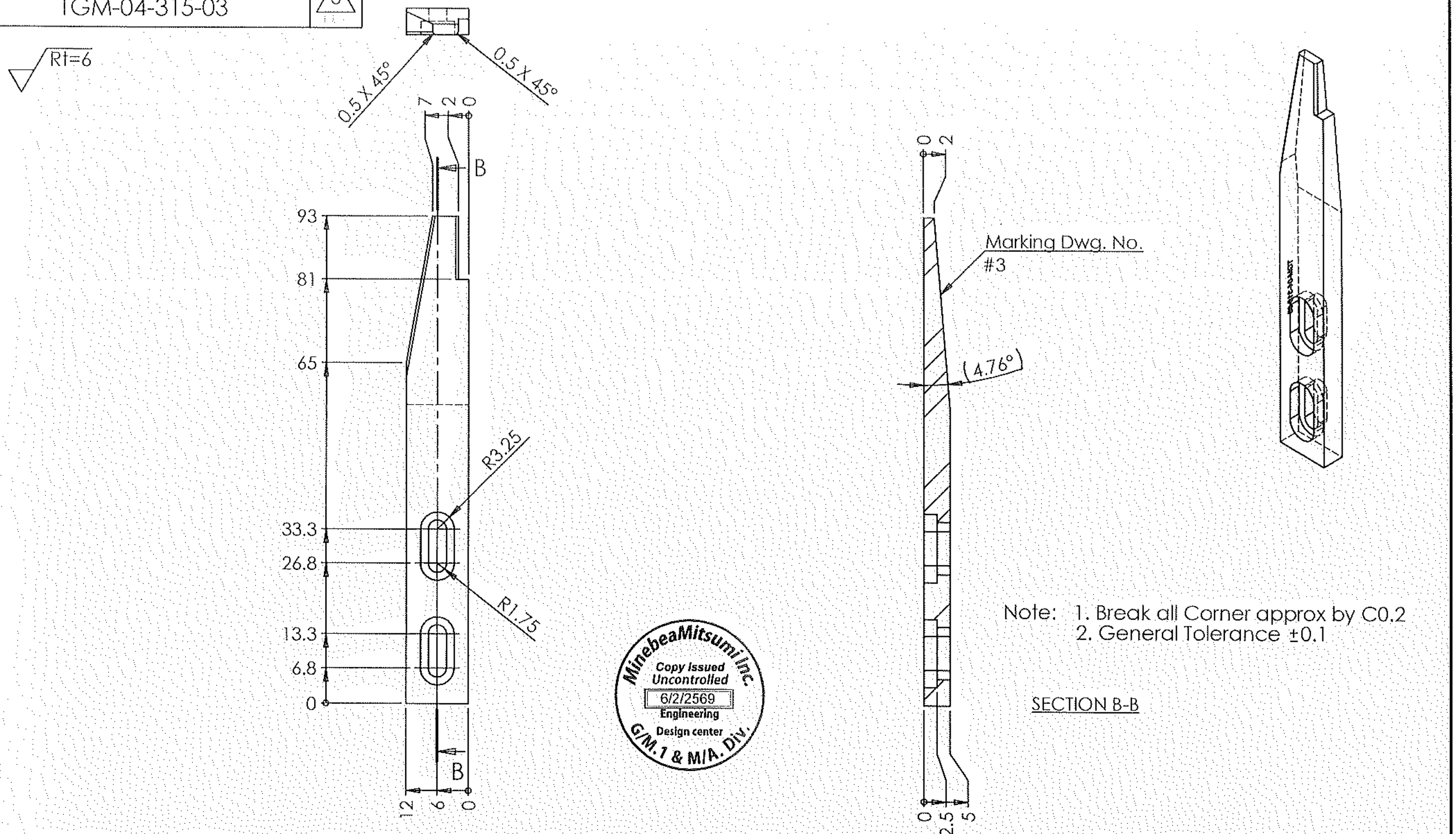
Vis Sleeve	1		Request No. 2015		3 RD ANGLE
MATERIAL Peek gray		SCALE	SHEET		UNIT (mm.)
FINISH -		DETAILED Phalachai	DESIGNED Aphisek	CHECKED Preecha	APPROVED Preecha
HEAT TREATMENT -		DATE 15 Feb' 12			

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Rail (Right)

PART No. TGM-04-315-04 | REV. 2

PART No.
TGM-04-315-03
3



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

SECTION B-B

MinebeaMitsumi Inc.
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G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	8 Aug'12	Preecha	Add: C'Bore
3	----	17 Aug'12	Preecha	Add: 65

GENERAL TOLERANCE	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.5
ANGLE	±1°

M/C: Vis Sleeve	Q'TY: 1
MATERIAL: Peek gray	
FINISH: -	
HEAT TREATMENT: -	

Dist. Request No.: 2015			3 RD ANGLE
SCALE	SHEET		UNIT (mm.)
DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha
DATE: 16 Feb' 12			

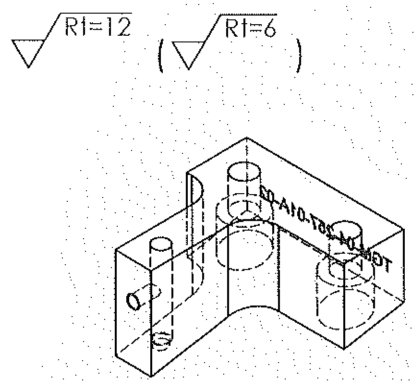
NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

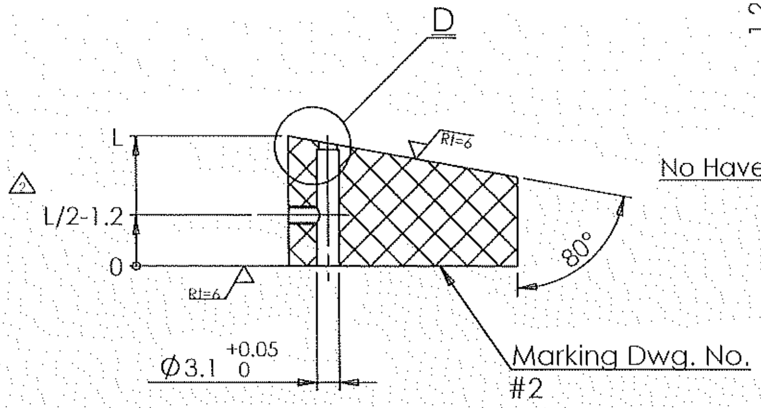
TITLE: Rail (Left)

PART No. TGM-04-315-03
3
REV.

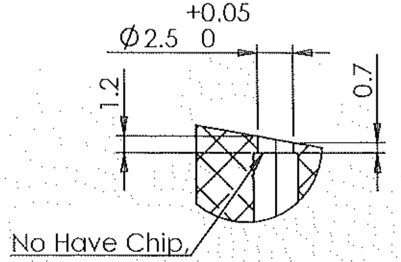
PART No. TGM-04-257-01A-xx 2

Item	W	L
-01	3.8	19
-02	9	19
-03	10	19
-04	12	19
-05		

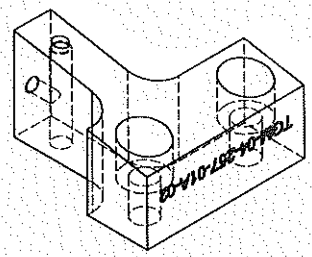


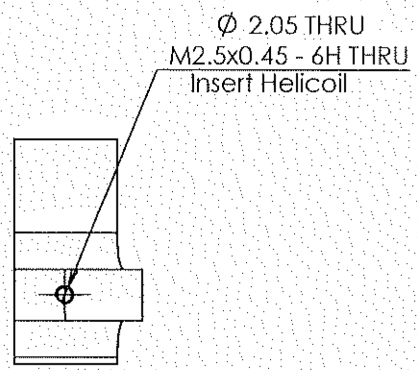


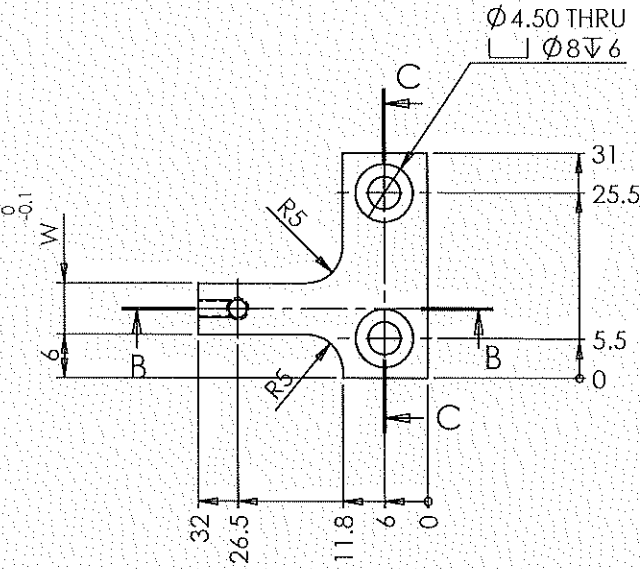
SECTION B-B
SCALE 1 : 1

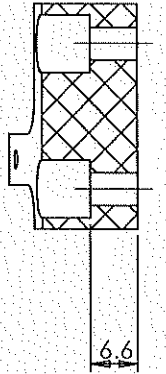


DETAIL D









SECTION C-C
SCALE 1 : 1

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Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

For: Fix lock Sensor No. FU-48 keyence

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	----	ENGINEERING RELEASE
2	----	6 Feb'26	Preecha	Change: 7 ==> L/2-1.2

[illegible]	±0.05
[illegible]	±0.1
[illegible]	±0.2
[illegible]	±0.3
[illegible]	±0.4
[illegible]	±0.5
ANGLE	±1°

M/C Vis Sleeve
Q'TY 1
MATERIAL Nylon Blue
FINISH -
HEAT TREATMENT -
List Request No. 439/16
3rd ANGLE
SCALE
SHEET
UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

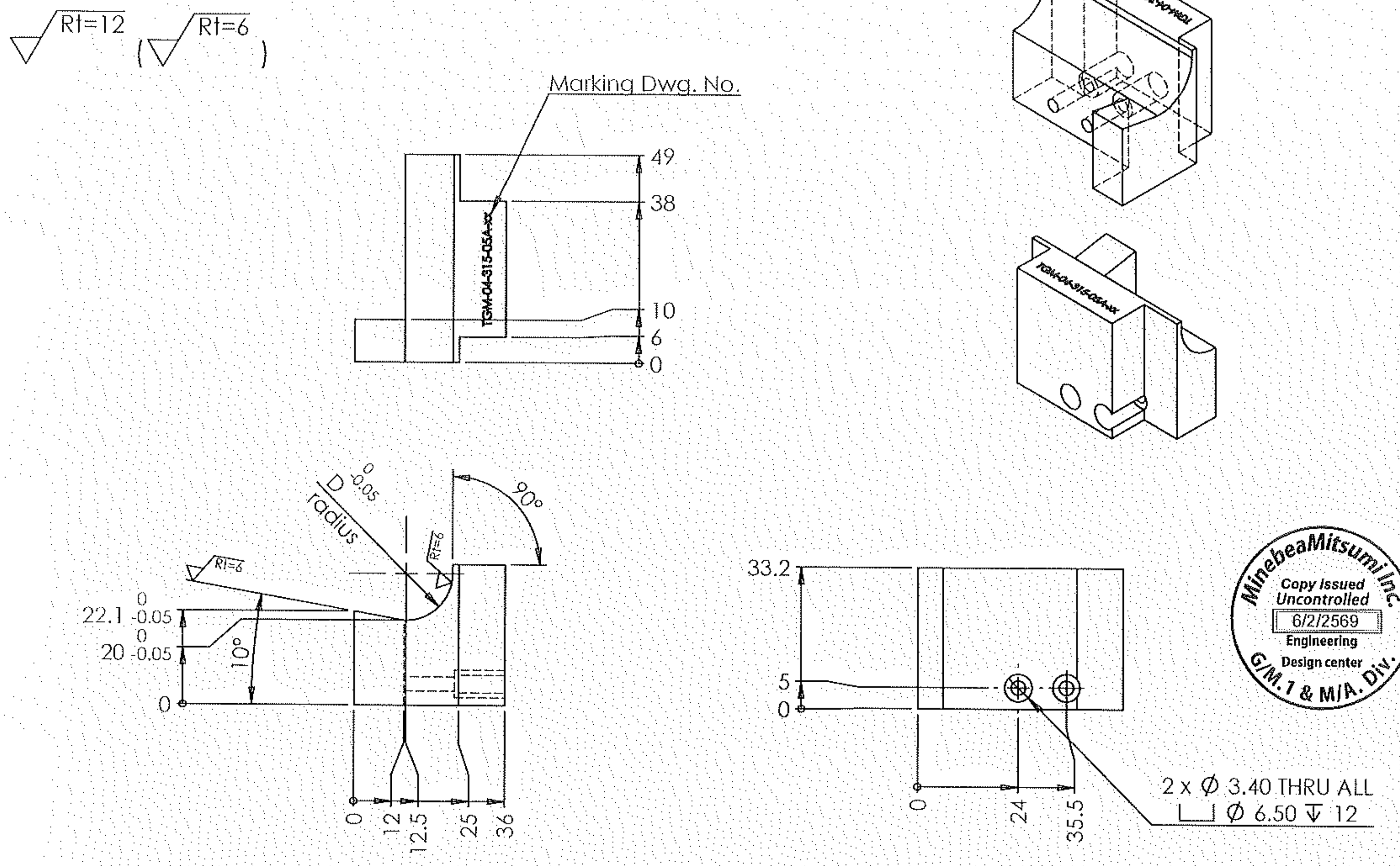
DATE 23 Sep' 16

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE Sensor Block (A)

PART No. TGM-04-257-01A-xx
REV. 2

PART No. TGM-04-315-05A-xx △1



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6/2/2569
Engineering
Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	D	For P/No.
-01	11	OT-7018-1
-02	15	OT-7069-G1B
-03		
-04		
-05		
-06		
-07		
-08		
-09		
-10		
-11		
-12		
-13		
-14		
-15		

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

[illegible]	±0.05
[illegible]	±0.1
[illegible]	±0.2
[illegible]	±0.3
[illegible]	±0.4
[illegible]	±0.5
ANGLE	±1°

M/C: Vis-SL
Q'ty: 1
MATERIAL: Nylon blue
FINISH: -
HEAT TREATMENT: -
Dist Request No.: 652/17
3RD ANGLE
SCALE
SHEET
UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE: 16 Nov' 17

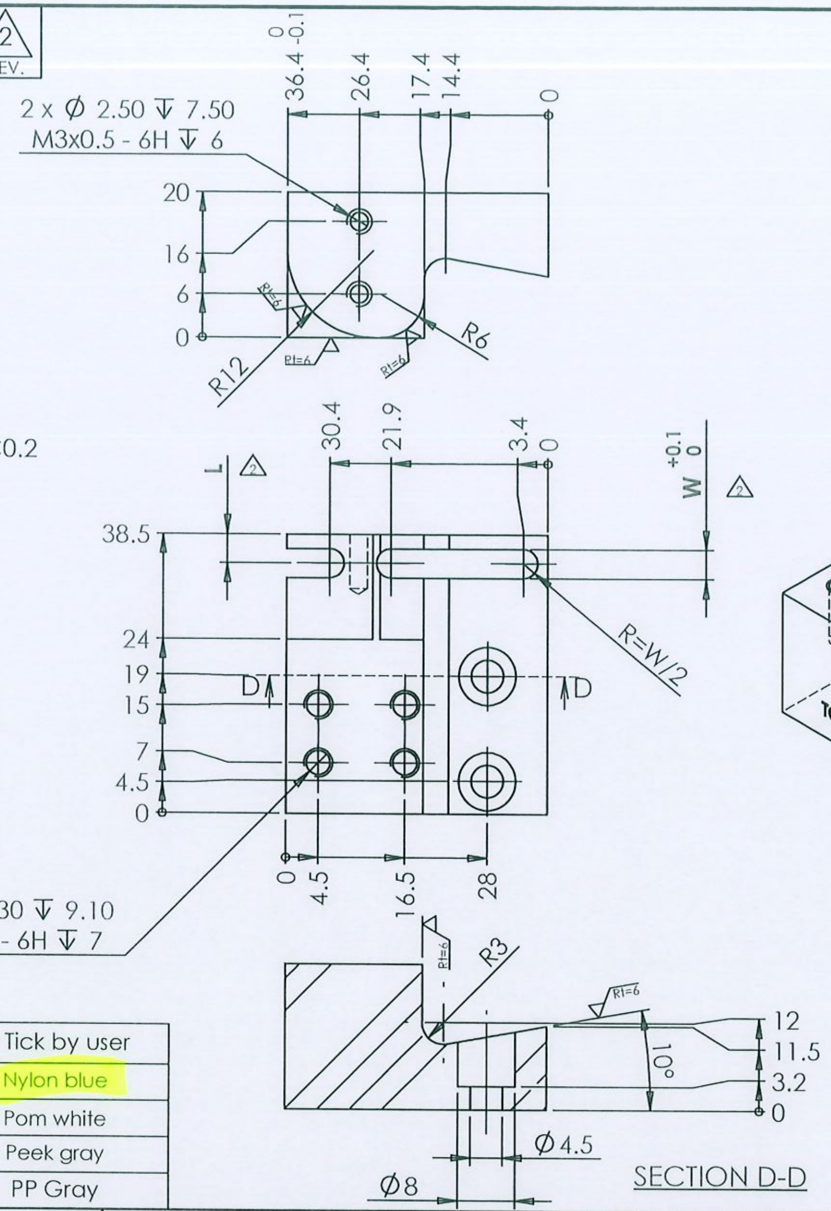
NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Guide Part (A)

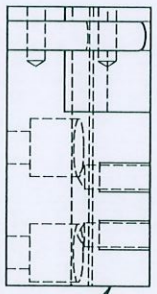
PART No. TGM-04-315-05A-xx △1 REV.

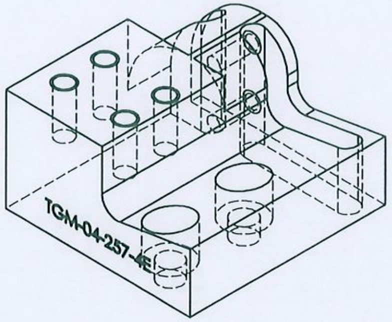
PART No.
TGM-04-257-4E-xx △2 | △2 REV.

√Rt=12 (√Rt=6)



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1





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Engineering
Design center
G/M.1 & M/A. Div.

☑ Mat'l Tick by user
☑ Nylon blue
☐ Pom white
☐ Peek gray
☐ PP Gray

△2 ITEM	L	W	Check
-01	4	4	Preecha
-02	5.3	5	Preecha
-03			
-04			
-05			
-06			
-07			
-08			
-09			
-10			
-11			
-12			
-13			
14			
-15			

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	5 Feb'26	Preecha	Change: TGM-04-257-4E ==> -257-4E-xx Etc.; RQ.352/25

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S
6.3S
25S
100S
∼ UNCUT

M/C. Vis Sleeve
Q'ty 1
MATERIAL See table
FINISH See table
HEAT TREATMENT -

List Request No. 652/17
SCALE :
SHEET /
3 RD ANGLE
UNIT (mm.)

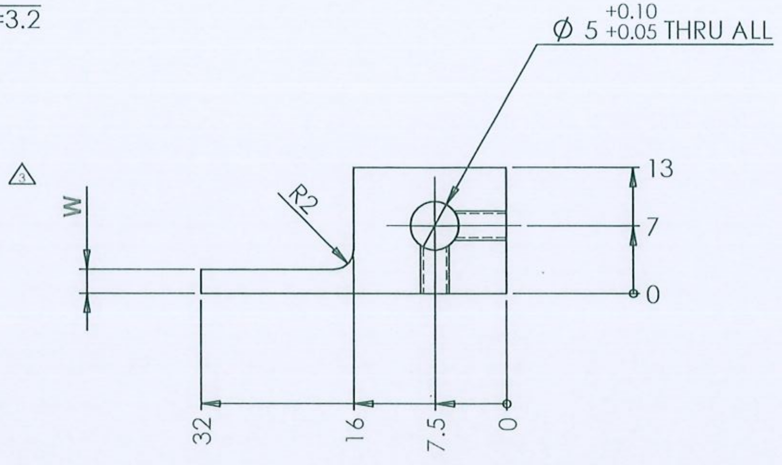
DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

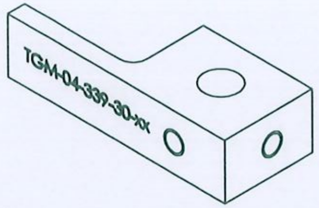
DATE 21 Nov' 17

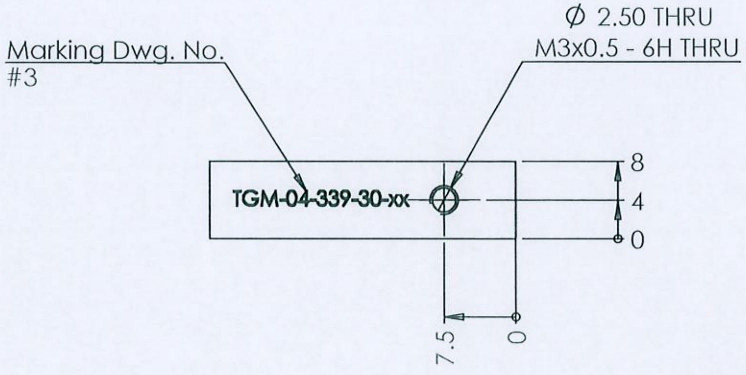
NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1
TITLE:
Rail end (E)
PART No.
TGM-04-257-4E-xx
△2 REV.

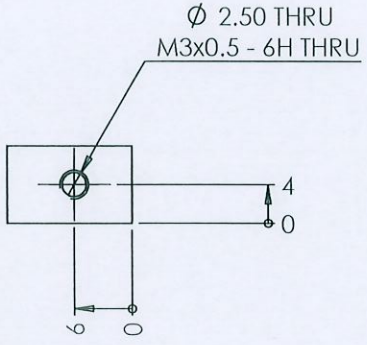
PART No.
TGM-04-339-30-xx
3
REV.

Rt=3.2









Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

ITEM	W	Check
-01	2.5	Preecha
-02	3	Preecha
-03	4	Preecha
-04	5	Preecha
-05		
-06		
-07		
-08		
-09		
-10		
-11		
-12		
-13		
14		
-15		

☐	Mat'l Tick by user
☑	Nylon blue
☐	Pom white
☐	Peek gray
☐	PP Gray

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Design center
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REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	18 Nov'14	Preecha	Change: M2.5 ==> M3 Etc.; RQ.676/14
3	----	5 Feb'26	Preecha	Change: 2.5 ==> W Etc.; RQ.352/25

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S
6.3S
25S
100S
UNCUT

M/C. Vis Sleeve	Q'ty 1
MATERIAL See table	
FINISH See table	
HEAT TREATMENT -	

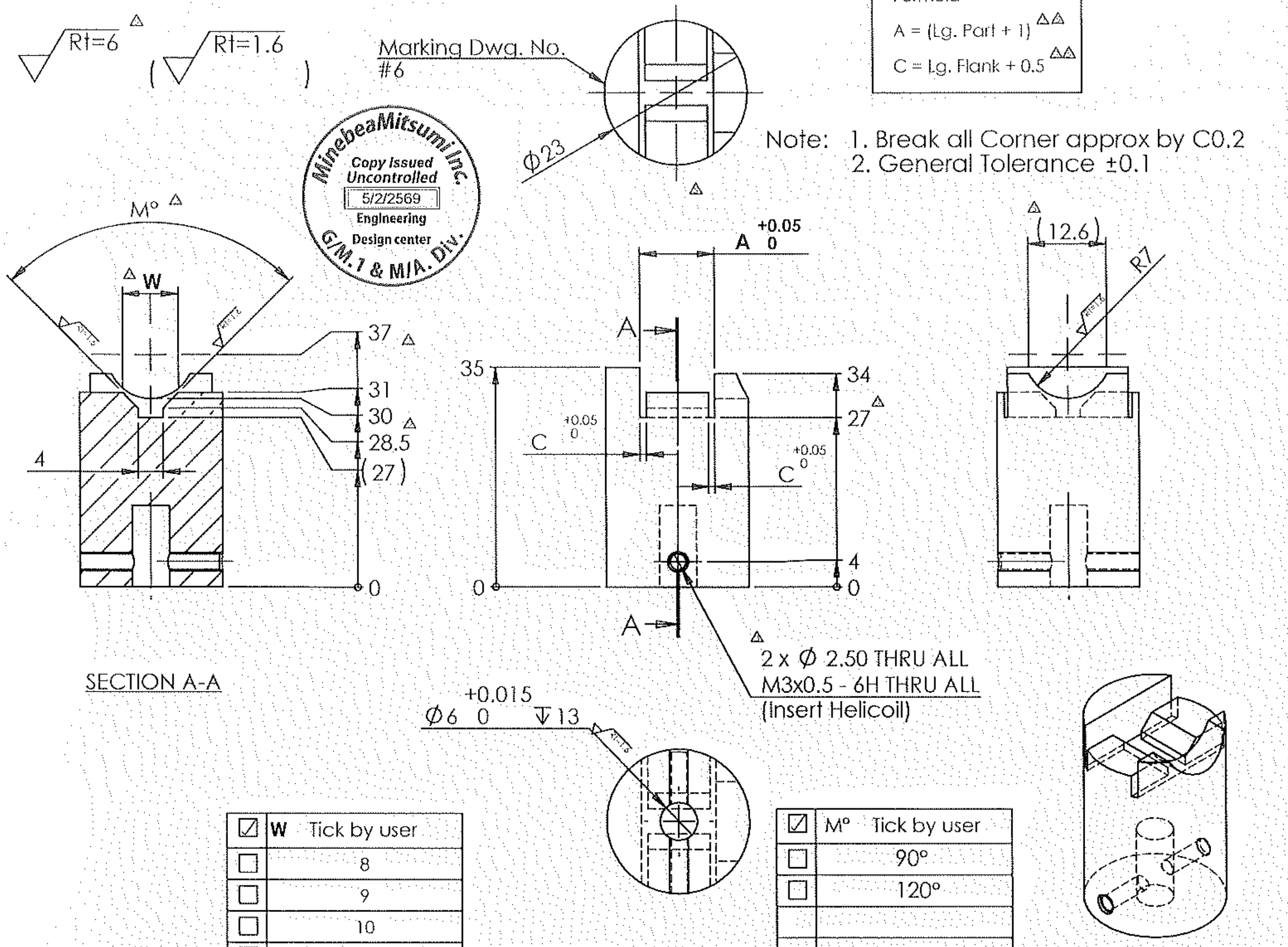
List Request No. 2780			3 RD ANGLE
SCALE :	SHEET /		UNIT (mm.)
DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha
DATE 29 Jan'13			

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE:
Pusher loader

PART No.
TGM-04-339-30-xx
3
REV.

PART No. TGM-04-257-3-xx 6



Formula
A = (Lg. Part + 1)
C = Lg. Flank + 0.5

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

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G/M.1 & M/A. Div.

☑	W Tick by user
☐	8
☐	9
☐	10
☐	11

☑	M° Tick by user
☐	90°
☐	120°

Item	A	C	Common total Lg. Part	For P/No.	Check
-01	16	1.6	15.596	AS-4356-1, AS-4289-1	Preecha
-02	7.5	1.6	7.09	-	Preecha
-03	6.8	1.6	6.35	-	Preecha
-04	9.3	1.6	8.89	-	Preecha
-05	5.8	1	-	AS-4381-1	Preecha
-06	7.8	1.4	-	F2002-1 AS-4373-1	Preecha
-07	10.34	2	-	AS-4307-1	Preecha
-08	11	1	-	AS-4390-1,T7069-G1B	Preecha
-09	18.7	1	-	F3002-1	Preecha
-10	15.5	1.6	14.7	A4339-1 A4344-1	Preecha
-11	20	1.6	13.1 to 19	-	Preecha Rq.90/17
-12	5.5	1	3 to 5	-	Preecha Rq.90/17
-13	13.3	1	5.1 to 13	-	Preecha Rq.90/17
-14	14	1	13.8	T7099AM1A	Preecha Rq.352/25
-15					

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	30 Jul'12	Preecha	Change: A=(Work L)+0.4 ==> A=(Lg. Part)+1 etc.;
3	----	15 May'13	Preecha	Change: M2.5 ==> M3
4	----	27 May'14	Preecha	Change: Item -07 (C) 2 Rq.314/14
5	----	22 Feb'17	Preecha	Cancle: D-Cut etc.; Rq.90/17
6	----	5 Feb'26	Preecha	Change: 9 ==> W etc.; Rq.352/25

	±0.05
	±0.1
	±0.2
	±0.3
	±0.4
	±0.5
ANGLE	±1°

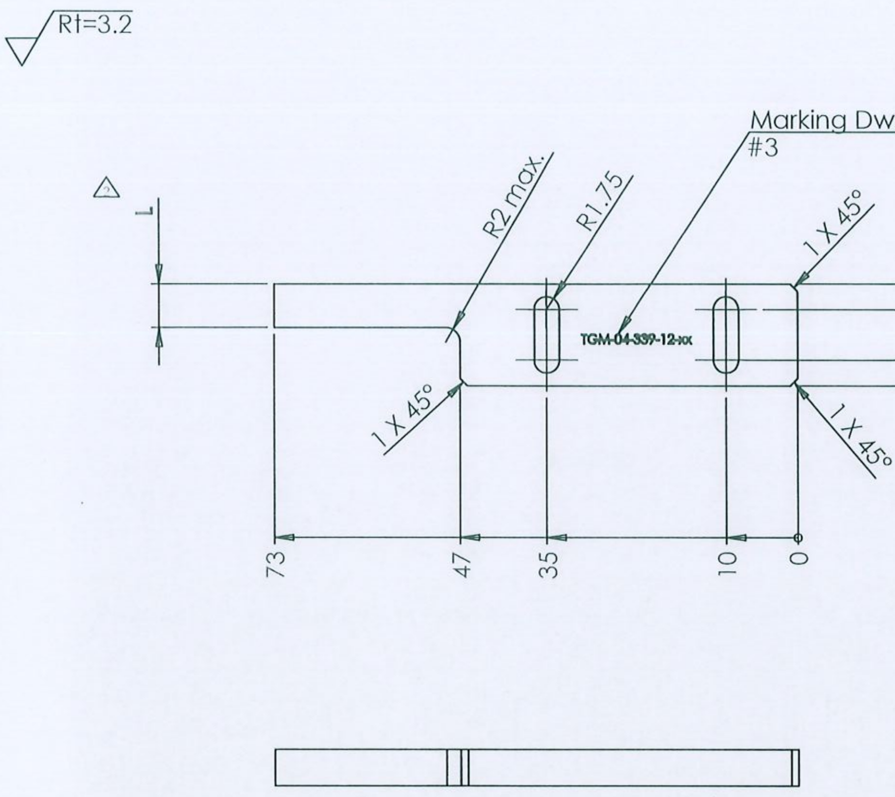
M/C: Visual Sleeve
Q'ty: 1
List Request No.: 2400
MATERIAL: Nylon Blue
FINISH: -
HEAT TREATMENT: -
3RD ANGLE
SCALE
SHEET
UNIT (mm.)
DETAILED: Phalacha
DESIGNED: Aphisek
CHECKED: Preecha
APPROVED: Preecha
DATE: 8 Oct' 10

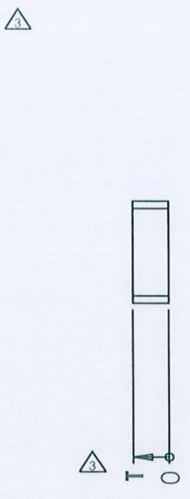
NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Turn Over Tool

PART No. TGM-04-257-3-xx
REV. 6

PART No. TGM-04-339-12-xx | 3 REV.





ITEM	L (2)	T	Use for Lg. Part	Check
-01	3	5	4 to 7	Preecha
-02	6	5	8 to 15	Preecha
-03	6	10	7.355	Preecha
-04	12	8	24 (OT7009)	Preecha
-05				
-06				
-07				
-08				
-09				
-10				
-11				
-12				
-13				
14				
-15				

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

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G/M.1 & M/A. Div.

☑	Mat'l Tick by user
☑	Nylon blue
☐	Pom white
☐	Peek gray
☐	PP Gray

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	31 Oct'13	Preecha	Change: 3 ==> L Etc.: RQ.3297
3	----	16 Nov'17	Preecha	Change: 5 ==> T Etc.: RQ.562/17

UNLESS SPECIFIED C0.1 OR R0.1

GENERAL TOLERANCE JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S
6.3S
25S
100S
~ UNCUT

M/C.	Vis. Sleeve
Q'ty	1
MATERIAL	See table
FINISH	See table
HEAT TREATMENT	-

List Request No. 2593

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 7 Nov' 13

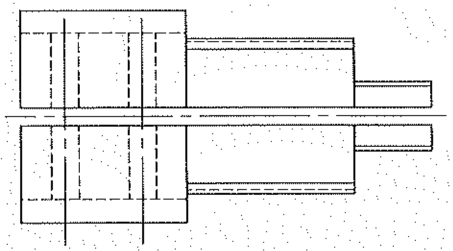
NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

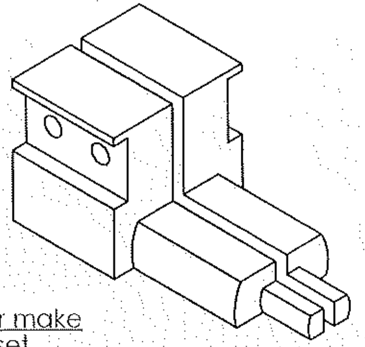
TITLE: Pusher

PART No. TGM-04-339-12-xx | 3 REV.

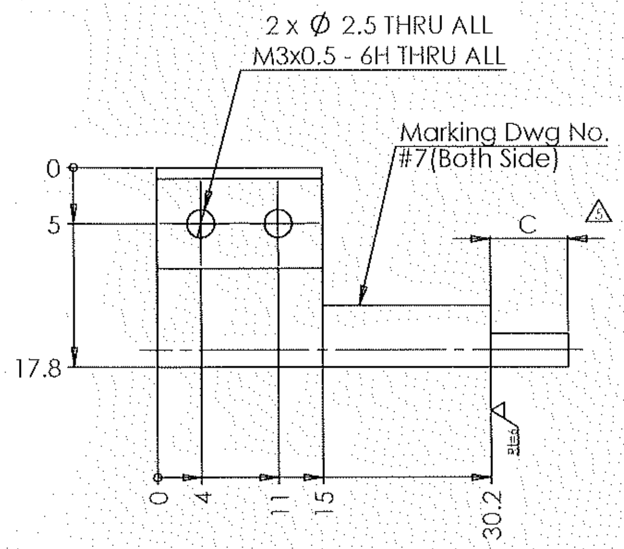
PART No. TGM-04-195B-xx △7

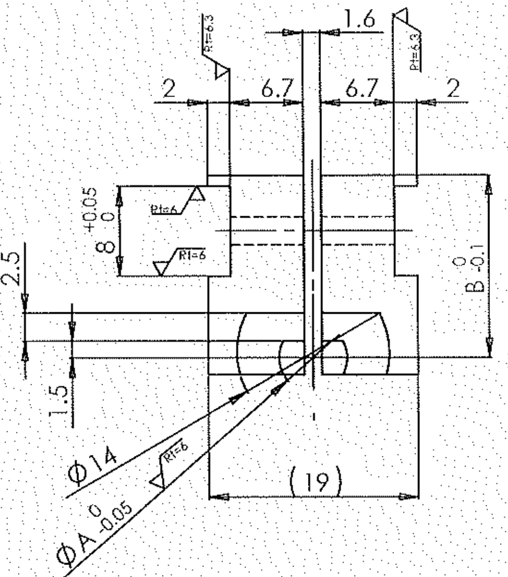
Rt=12 (Rt=6)





For: Vendor make
2 pcs./set





Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	A	B △4	C △5	For ID.	Check
△2 -01	8	15.3	2.5	9.525	Preecha
-02	6	16.3	2.5	7	Preecha
103	6	16.3	5	T7164-G1C	Preecha
-04	10	16.3	3.5	OT-7099A	Preecha
-05	6	16.3	7	OT-7069	Preecha
-06					
-07					
-08					
-09					
-10					
-11					
-12					
-13					
-14					
-15					

MinebeaMitsumi Inc.
Copy Issued
Uncontrolled
5/2/2569
Engineering
Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	----	ENGINEERING RELEASE

UNLESS SPECIFIED
SCALE OF ...
GENERAL TOLERANCE
ECONOMIC MACHINING
WORKING

0.5 THRU 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

M/C: Vis Sleeve
QTY: 2
MATERIAL: Peek gray
FINISH: -
HEAT TREATMENT: -

Mfg. Request No. 359/15
3 RD ANGLE
SCALE
SHEET
UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Aphisek	Aphisek	Preecha	Preecha

DATE 15 Oct'15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Transfer Chuck (B)

PART No. TGM-04-195B-xx △7